

# SARTHAK SETHI

+1-548-398-7610    s36sethi@uwaterloo.ca    LinkedIn    GitHub

## EDUCATION

University of Waterloo  
Honours Bachelor of Computer Science

Waterloo, ON  
Sep 2024 – May 2029

## TECHNICAL SKILLS

**Languages:** Python, C/C++, Bash, TypeScript, JavaScript, SQL, HTML/CSS  
**Frameworks:** FastAPI, React, Next.js, Node.js/Express, LangChain/LangGraph, WebSockets, OpenAI API, Azure OpenAI, Anthropic API, Tailwind CSS  
**Cloud & Infra:** Azure (AKS, Container Apps), AWS (EC2, S3), Kubernetes, Istio, Docker, GitHub Actions, CI/CD, Helm  
**Data & Tools:** PostgreSQL, Supabase (RLS), Redis, Temporal, Alembic, SQLAlchemy, Keycloak, Azure AD/MSAL, Pandas, Git, Playwright

## WORK EXPERIENCE

- AI Software Engineer & Project Manager, CSI — Volaris Group** Jan 2026 – Aug 2026  
Mississauga, ON
- Delivered **4 enterprise products** reporting directly to the **CTO** — ran client discovery meetings, shipped **skeleton prototypes** to validate requirements, and owned architecture through production across React/TypeScript, Python/FastAPI, and Azure (**820+ PRs**).
  - Built an internal **AI Coding Agent** adopted by **14,000+ developers** across CSI — autonomously detects issues, generates specs (human-approved), implements fixes, runs **Playwright** tests on staging, triggers **code review** for security/infra gaps, validates **CI checks**, and requires human approval before merge to main.
  - Sole engineer/PM for **Canvas LMS on Azure Container Apps** with **PITR recovery** and zero-downtime deploys; architected a **Legal Document Analysis** platform on **AKS** (Istio, Trivy) with AI-powered **OCR**, **contract risk assessment**, **RBAC**, and hybrid **SSO** — **240+ PRs**.
  - Built an **AI portfolio risk engine** across **11 dimensions** with **Temporal durable workflows** and a real-time **Speech AI** product using **Azure OpenAI Realtime API** that cut voice-response latency from **10–15s to <3s** — **580+ PRs**.
- Software Developer, Danier** Jun 2025 – Aug 2025  
Toronto, ON
- Built a **Low-Stock Alert System** using FastAPI, SQLAlchemy, and **Pandas/OpenPyXL**, sending automated **HTML email alerts** to stakeholders and improving on-page **SEO** to boost search rankings.
  - Developed a custom **AI chatbot** with **FastAPI**, React, and Tailwind CSS, integrating FAQ/product search, analytics hooks, and caching for faster and more reliable responses across the site.
- Software Engineer, Nokia** Jun 2023 – Jul 2023  
India
- Modernized bug tracking with Node.js/Express and optimized **PostgreSQL** queries, reducing load times by **47%** and improving database query performance for internal tools.
  - Created React dashboards with **D3.js** visualizations, containerized with **Docker** for deployment and consistent results across developer environments.

## PROJECTS

- Valldor - Procurement AGI** | Next.js, FastAPI, LangChain, PostgreSQL, Three.js Waterloo, ON  
Selected for a **Y Combinator interview** and received a **conditional offer** from **Andreessen Horowitz (a16z)** for this autonomous procurement platform.
- Built an **autonomous Procurement AGI** handling the full lifecycle: ERP monitoring → vendor discovery → RFQ → **multi-round AI negotiation** → PO generation → logistics tracking → 3-way match → payment.
  - Engineered a **global landed-cost engine** computing duty, freight, and trade-agreement preferences for any delivery point worldwide, with a **4-layer market intelligence system** scoring vendors across 6 axes.
  - Designed a **multi-tenant SaaS** with buyer/seller workflows, onboarding wizard, ERP purchase orders, real-time AI learning from vendor patterns, and a **3D AI avatar** (Three.js + Web Speech API) for voice-driven procurement.
- Muse Sketch Studio** | React, TypeScript, Node.js/Express, Replicate API, Tailwind CSS Toronto, ON  
Won **1st Place** at the **Replicate AI Hackathon (2025)** among **100+ participants** (\$1,000 cash + \$500 Amazon gift card).  
Built an AI design pipeline (**prompt** → **sketch** → **colorization** → **video**) with React/TypeScript frontend, Node.js backend, and Replicate API (google/veo-3) for **PNG/MP4** export.
- Danier Bot** | FastAPI, React, OpenAI API, SQLAlchemy, Pandas Toronto, ON  
Built **Danier Bot**, a unified **AI assistant for customers and employees** that answered product questions from auto-synced **Shopify data**, supported product discovery, monitored inventory continuously, and sent automated **HTML low-stock alerts** to internal teams — deployed on **Render** with **100% alert reliability**.
- CarRaksha — Safe Driving System** | C++, Arduino India  
Won **1st Prize** at the **All India Techfest Hackathon (100,000+ students)**. Built a collision-prevention system with Arduino, ultrasonic/MQ-3 sensors, ignition lockout, and speed-limiting — **79% reliability** across 10+ vehicle models.